

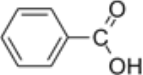
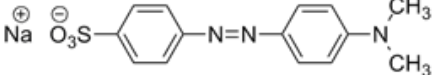
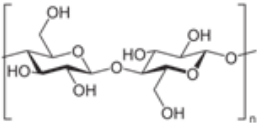
Activity 3: Extraction and Recrystallization: Separation and Purification of Benzoic Acid

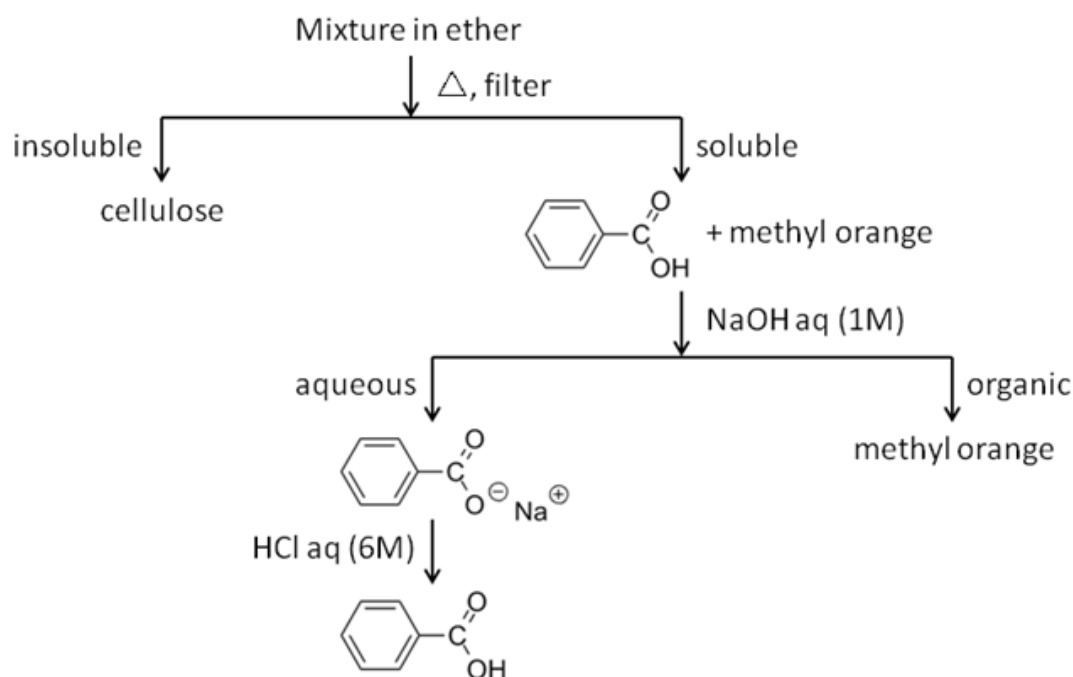


Figure 1. Benzoic acid is a colorless crystalline solid. [1]

Activity 3: Procedure

The Experiment

benzoic acid	methyl orange (turns pink color in low pH solution)	cellulose
		



Lab Activity Goal

Use extraction (solid-liquid and chemically active), along with other organic laboratory techniques, to separate benzoic acid from a mixture containing cellulose (a natural polymer of glucose) and methyl orange (a common acid/base indicator). Once the benzoic acid (impure/crude) is isolated from the mixture it will be purified by recrystallization from water. During the next lab, the crude and pure recrystallized benzoic acid will be characterized by melting point to assess relative purity.

Activity 3: Procedure

Pre-Laboratory Assignments

- See the Activity 3 Pre-Lab Preparation and Outline document in the Activity 3 Module on Labflow for detailed directions on what you need to do BEFORE coming to lab.
- Remember, you will need to be properly dressed (see the Safety Instructions you signed last week on Labflow). You must bring your flame-resistant lab coat, splash goggles, and face mask to do Activity 3.

ALERT: Laboratory Techniques

- In Part 1, you will be using a separatory funnel and steam bath. Pay careful attention to your instructor's directions to safely use and vent your separatory funnel. Also, pay careful attention to your instructor's directions to avoid **steam burns** when using the steam bath.
- In Part 2, you will be using a heating mantle and performing hot gravity filtration. Pay careful attention to directions on how to safely set up a heating mantle.

Experimental Procedure

Part 1. Separation of Benzoic Acid from a Mixture

1. Use your prepared lab notebook pages to record what you measure and observe during the entire experiment. You will be uploading images of your completed notebook pages to a Labflow tile before you leave lab.
2. Ensure that the steam bath in your hood is empty of water (condensed from a previous lab period). If there is standing water in the bottom of the bath, drain it by lifting and tipping the bath so that the water can drain through the red hose directly into the cup sink in the back of your hood. This proactively prevents flooding of the bath during your experiment. **Once empty, place the steam bath on top of a lowered lab jack, rather than leaving it on the bench top. This helps condensed water to drain into the cup sink (water runs downhill).**

Activity 3: Procedure

3. Weigh ~4 g of the crude benzoic acid mixture using the weighing balance and place it in a 125-mL Erlenmeyer flask. Record into your notebook the exact mass that you measured.

- **Note:** You may need to use a watch glass in place of the weighing paper/boat. It is not important that you use exactly 4 g of crude mixture, but it *is* important that you record (in your notebook) the weight that you actually used.

4. Add 50 mL of diethyl ether to the 125-mL Erlenmeyer flask.

- **Note:** Ether is a volatile (high vapor pressure) and flammable solvent. It should only be handled inside hood spaces to prevent the laboratory air from becoming *too fragrant*.

5 Add 2-3 boiling stones and then heat the mixture on a steam bath until the ether begins to boil. To prevent boiling over, place the flask to the side of the steam bath rather than directly over the center hole.



Important: Safety Pause

The boiling stones will prevent bumping. Add the stones **before** you begin heating the liquid.

Safety Pause

6. Remove the **ether insoluble** material from the ether solution using the technique of Büchner funnel vacuum filtration.

- **Think About This:** What is the ether insoluble material trapped in the Büchner funnel? Why doesn't it dissolve in ether? What chemicals are dissolved in the ether? Why did they dissolve in the ether?

7. In preparation for **Part 2** your instructor may direct you to begin heating water using the steam bath. Add about 200 mL of water to a 250-mL round-bottomed flask with two or three boiling stones. Secure this flask to the monkey bars using a clamp. Do this on the left or right edge of the hood space so it is out of the way.

Activity 3: Procedure

8. Place the filtrate (ether solution) from step 6 in a 500-mL separatory funnel. Rinse the filter flask with 15-25 mL of ether (based upon your estimate of how much ether evaporated during steps 5 and 6) and transfer to the separatory funnel. Finally, add 30 mL of 1 M NaOH. After venting, swirl gently a few times while venting often. Place the funnel back into the iron ring. Remove the cap and allow the layers to separate.

- **Note:** Follow the extraction/venting procedures discussed in the lab technique videos on Labflow (“Using a Separatory Funnel” and “Working Up Reactions with Washes and Extractions”).



Important: Safety Pause

Always use and vent the separatory funnel in the hood, pointed away from others. Make sure to hold the stopper securely when mixing the contents of your funnel to prevent chemical spills.

Safety Pause

- **Think About and Sketch into your Notebook:** Sketch what you see in the funnel. What layer is the ether solution? What layer is the aqueous layer? Label both layers. What chemical reaction happened when the NaOH(aq) came into contact with the ether solution? What chemicals are favorably partitioning into each layer? Which layer has what you want?
9. Drain the aqueous layer into a 250-mL beaker or flask. **Label this beaker.** Extract the ether layer again with another 30 mL of 1 M NaOH. Drain the aqueous layer into the flask that already has the first aqueous extraction.
- **Note:** Remember to always follow proper extraction procedures. Vent often into the hood!
10. Cool the combined aqueous extraction solutions (in the 250-mL beaker/flask) on ice. Acidify the solution with 50 mL of 6 M HCl (which was also cooled on ice). Note: Your 6 M HCl flask should be labeled.

Activity 3: Procedure

- **Observe and Record in your Notebook:** What happened? What is the solid? Provide the chemical reaction for the solid formation. Why did the solid form? Why do you think the solution turned pink?

11. Use a Büchner funnel to collect the solid precipitate by vacuum filtration.

Note regarding steps 8-11: In the future, you will be more experienced working in an organic laboratory. For more experienced students, steps 8 through 11 would be combined into one step and simply would say: Extract the ether filtrate with two 30 mL portions of 1 M NaOH. Cool the combined aqueous extractions in a 250-mL flask on ice and carefully acidify with 50 mL of cold 6 M HCl. Collect the precipitated crystals using vacuum filtration.

12. Place your crystals collected in the Büchner funnel on a tared (pre-weighed) watch glass. Transfer a very small portion of these crude crystals to a small, labeled beaker in your drawer for a melting point experiment next week. Weigh the remaining crystals to be used in Part 2.

Part 2. Recrystallization of Crude Benzoic Acid

1. Continue to use your lab notebook pages to record what you do, measure, and observe during this experiment. Notebooks that are useful allow you to reconstruct what occurred in lab.
2. Place your crude benzoic acid crystals into a 250-mL Erlenmeyer flask. Remember to save (in a small beaker) a small portion (petite pea size) of the crude crystals for next week!



Important: Safety Pause

CAUTION – FIRE HAZARD: Be sure that no ethyl ether remains in your hood before proceeding to the next step. Your lab instructor will increase the exhaust rate of your hood to ensure all ether is removed before you can proceed with the next step.

Safety Pause

Activity 3: Procedure

3. Transfer the pre-heated water on your steam bath from step 7 in Part 1 to a heating mantle. Use the clamp as a handle and wear the large, heat-resistant gloves. Heat the water to a boil using the heating mantle. **While waiting for the water to boil, ensure that you have cleaned up all the glassware from the Part 1 activities.**



Important: Safety Pause

CAUTION: Never plug the heating mantle directly into an electrical outlet. Only plug it into the Variac located below the hood sash.

Safety Pause

4. Slowly add the boiling water to the crude crystals using a clamp as a handle (secured properly) and oven mitts as extra protection. The idea is to get the crystals to dissolve in a **minimum** amount of boiling water. Why?

- Your crude crystals may contain insoluble impurities like specs of filter paper. Don't try to dissolve these insoluble specs! If you try to dissolve them, you will end up adding too much water and your crystals will never precipitate out of the cooled recrystallization solution.
- It is important that the crystal solution be kept at or near boiling to maximize crystal solubility. Thus, keep the crystal solution mixture on a steam bath.
- You may need to use a clamp as a handle for swirling, and later for pouring. If you do use a clamp it needs to be attached to the monkey bars when it is on the steam bath to prevent tipping. If you don't use a clamp you must use oven mitts. Oven mitts are still recommended when using the clamp as a handle.

Your solution may be light pink, or it may even be dark pink. Why? Pink solutions need to be decolorized using charcoal.

- The charcoal powder is finely divided carbon. The carbon adsorbs organic molecules; especially compounds that are highly conjugated (a common property of colored chemicals).

Activity 3: Procedure

- Remove the pink solution from the steam bath and swirl to dissipate any hot spots in the solution. This will prevent foaming upon addition of the charcoal.
- Add a very small amount (small pea size or less) of charcoal, swirl, and immediately return to the steam bath to prevent crystals from forming.
- You should observe pink coloring begin to disappear. If it is still strongly pink, add a tiny bit more of the charcoal.
- Clean up any mess you made handling or spilling the charcoal. (Safety & Citizenship points!)

5. Now you are ready to perform a hot gravity filtration. Remember what you learned when you watched the “Performing a Hot Gravity Filtration” video on labflow.

- Remember to use fluted filter paper. Set up the funnel, fluted filter paper and flask apparatus.
- Wet the filter paper with some of the fresh hot water and then begin to carefully pour your benzoic acid solution into the funnel. Add it slowly enough not to overflow but add it fast enough to never let the filter paper dry out. If precipitate starts forming in the filter paper, then get new paper.
- Your filtered solution should be very clear. Note that clear does not imply colorless.

6. Remove the filtrate from the steam bath and allow it to cool to room temperature in your hood. Why do you want the solution to cool slowly?

- While you are waiting for your crystals to form, ensure that you have cleaned up all the glassware and space as much as possible. Ask your instructor if there is something else that can be cleaned.

7. As the solution begins to cool you should see crystals begin to form. If the solution does not produce crystals by the time it reaches room temperature you will need to try to artificially nucleate the solution. Your instructor will discuss these methods with you.

8. If you still don't get crystals, then cool your solution on ice. If you still don't get crystals, then you added too much water during the dissolving process!

Activity 3: Procedure

Ask your instructor for some advice and be sure to record your problem in your notebook.

9. Collect the crystals by Büchner funnel vacuum filtration.

10. Transfer your purified crystals to a tared (pre-weighed) beaker and carefully place them in your drawer until next week. Break the crystals up so that they have maximum surface area to dry.

11. Use acetone first and then glass soap to clean your glassware and the community 500-mL separatory funnels.

12. Next week, you will immediately weigh your crystals (and beaker) and calculate the mass of your pure crystals. This allows you to calculate the percent composition of your original mixture.

13. Before you leave lab, upload your in-lab notebook pages to Labflow. Your in-lab notebook pages must include the following in a **single pdf file**:

- Any experimental observations (including but not limited to: color changes, observational notes, measurements, etc.)
- Your handwritten answers to the questions posted in the In-Lab Notebook Pages tile on Labflow
- A picture of your Organic Lab Cleaning Checklist

Clean-Up and Waste Disposal

Activity 3 Specific Cleanup

DO NOT THROW AWAY your crude or pure benzoic acid! You will need both next week for melting point determination and analysis.

- **Organic Liquid Waste**
 - Ether solutions
 - Acetone washes from glassware washing
- **Aqueous Waste**
 - Any leftover aqueous HCl
 - Any leftover aqueous NaOH
 - **Glass soap rinses from glassware washing**
- **Solid Waste (Red Can)**
 - Cellulose
 - Gloves
 - Filter paper
 - Do **NOT** drop filter paper into the liquid waste filter funnel. This prevents messy floods. Do **NOT** leave filter paper in the sink. This is poor lab etiquette.
 - Excess charcoal
 - Pipettes
- **Regular Waste**
 - Paper towels

Activity 3: Procedure

Routine Cleanup

- Clean up your hood space of all trash, spilled chemicals, etc. Don't leave a mess for the people who share this hood at other time slots. Up to 24 other people are sharing this space at various time slots throughout the week. Clean up your mess!
- No dirty or clean glassware should be left in the sink. No used filter paper or wadded up dirty paper towels should be left in the sink.
- Dirty paper towels go into the REGULAR TRASH not the red solid waste trash. This saves the university a lot of waste disposal money!
- Dispose of dirty filter paper in the red solid waste container.
 - Do NOT deposit filter paper in the liquid waste eco-funnel as this will cause a clog which will lead to a messy overflow of waste.
 - Do NOT leave filter paper in the sink as this is poor lab etiquette and rude.
 - Used gloves should be disposed of into the red solid waste.
- Ask your instructor if you are in charge of cleaning up the balance area for this week. If you are in charge do the following:
 - Be sure all reagent bottles are properly capped.
 - Use the small brush to clean off balances.
 - Use small broom and dustpan (located on the wall surrounding the balance area) to remove excess solid from counter. Dispose of excess solids in red solid waste container. Return broom and dustpan to proper location.
 - Wipe down the counter with a moist paper towel.
- Ask your instructor if you are in charge of making sure the sink area is presentable for the next lab group.
 - Wash any dirty glassware rudely left in the sink. Let clean glassware air dry on rack or windowsill area.
 - Dispose of any paper towels, filter paper, or cotton rudely left in the sink.
 - Do not remove broken glass from the drain or the sink. Rather, tell your instructor and then fill out a "something wrong form" located in the cabinet above the solid waste containers. Be sure to record room number and the Left/Right location of the sink.

Activity 3: Procedure

Post-Lab Assessment

The Activity 3 Post-Lab Assessment is due in 2 weeks. See deadline posted on Labflow.

Image Credits

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